Dynamo db. with AWS

Total Duration: 20 Days
AWS Services
Duration (36 Hours)
4 Hours per day for 9 Half Days

Topics	Sub Topics	Duration (hrs.)
Introduction to AWS	DC vs Cloud	**4 Hours (1st Day)**
	Features, Advantages	
	AWS Services overview	
	AWS Organizations	
Identity and Access Management	ID creation and management	
Monitoring in AWS	Cloud Watch intro	
	Billing alarms	
	AWS CLI	
AWS Storage	S3 Basics	**8 Hours (2nd & 3rd Days)**
	S3 Buckets creation	
	S3 Pricing Tiers	
	Security and Encryption	
	Version Control	
	Lifecycle Management and Glacier	
	S3 Lock Policies	
	Glacier Vault Lock	
	S3 Performance	
	S3 Select and Glacier Select	
	Sharing S3 between accounts	
	Cloud Front	
	Snowball Overview	
AWS Compute	EC2 Basics	**8 Hours (4th & 5th Days)**
	Creating EC2 instances and billing summary	
	Security Groups Basics	
	EBS	
	AMI Types (EBS and Instance Store)	
AWS Database	Introduction to AWS database and DAAS	
	RDS instances	
	RDS Backup, Multi AZ read replicas	
	Dynamo DB Introduction and details	
	Aurora	
	Elasticache	
AWS security basics	IAM basics and use to identity and access management	**4 Hours (6th & 7th Day)**
	IAM	
AWS Networking - Route 53	Basic of S3 and working	
	Basic of S3 and working	
AWS Networking - VPC	Basics Concepts	
	Access Control Lists (ACL)	
	Network Address Translation (NAT)	
AWS High Availability Architecture	Load balancer introduction and types	**4 Hours (8th Day)**
	Load balancer creation and health checks	

	Understanding, Creating and managing autoscaling groups	
Introduction of App Services	SQS (Queue concepts)	
	SWF	
	SNS	
Server Less	Lambda Concepts and Services	**4 Hours (9th Day)**
ETL	AWS EMS service	
	AWS Glue, Crawler	
	AWS Athena, AWS QuckSight, Data Visualization	

Dynamo DB
Duration (44 Hours)
4 Hours per day for 11 Half Days

Introduction	• Basics of Relational Databases • Data Normalization • Basics of NoSQL Databases • Types of NoSQL Databases	**4 Hours (1st Day)**
DynamoDB Basics	• Overview of AWS DynamoDB • Terminology Comparison with SQL • DynamoDB Tables and Naming Conventions • Data Types in DynamoDB • DynamoDB Consistency Model • DynamoDB Capacity Units • DynamoDB On-Demand Capacity • Basics of DynamoDB Partitions • Basics of DynamoDB Indexes • Local Secondary Indexes and Global Secondary Indexes • Interacting with DynamoDB	
Setting up DynamoDB	• Deploying • Usage notes • Setting up DynamoDB (web service)	**4 Hours (2nd Day)**
Accessing DynamoDB	• Using the console • Using the AWS CLI • Using the API • Using the NoSQL workbench • IP address ranges • Getting started with DynamoDB	**4 Hours (3rd Day)**
Create/Write/Read/Update/Query	• Prerequisites • Cheat sheet • Step 1: Create a table • Step 2: Write data • Step 3: Read data • Step 4: Update data • Step 5: Query data • Step 6: Create a global secondary index	**4 Hours (4th & 5th Day)**

	• Step 7: Query the global secondary index	

In Memory Acceleration with DAX	• How it works
	• DAX cluster components
	• Creating a DAX cluster
	  ○ Using the AWS CLI
	  ○ Using the console
	• Consistency models
	• Developing with the DAX client
	  ○ Tutorial: Running a sample application
	• Modifying an existing application to use DAX
	• Managing DAX clusters
	• Monitoring DAX
	  ○ Monitoring tools
	  ○ Monitoring with CloudWatch
	  ○ Logging DAX operations using AWS CloudTrail
	• DAX T3/T2 burstable instances
	• DAX access control
	• DAX encryption at rest
	• DAX encryption in transit
	• Using service-linked roles for DAX
	• Accessing DAX across AWS accounts
	• DAX cluster sizing guide
	• API reference
		4 Hours (6th Day)

Integration of DynamoDB with EMR and S3	• Configuring AWS credentials in your files using Amazon Cognito • Loading data from DynamoDB into Amazon Redshift • Processing DynamoDB data with Apache Hive on Amazon EMR • Integrating with Amazon S3	**4 Hours (7th Day)**
Autoscaling, Security, Streams & Migration	• What is Application Auto Scaling? • Setting up • Getting started • Services that work with Application Auto Scaling • Scheduled scaling • Target tracking scaling policies • Step scaling policies • Tutorial: Configuring auto scaling to handle a heavy workload • Suspending scaling • Monitoring	**4 Hours (8th Day)**

	• Security
	  ○ VPC endpoints (AWS PrivateLink)
	  ○ Data protection
	• Streams
	• Backup & restore / Disaster recovery for Pooled model / Archival mechanism
	• Encryption at rest
	• Migration from on prem to cloud	

Best Practices	• NoSQL design for DynamoDB
	  ○ Differences between relational data design and NoSQL
	  ○ Two key concepts for NoSQL design
	  ○ Approaching NoSQL design
	• Best practices for designing and using partition keys effectively
	• Using burst capacity effectively
	• Understanding DynamoDB adaptive capacity
	  ○ Boost throughput capacity to high-traffic partitions
	  ○ Isolate frequently accessed items
	• Designing partition keys to distribute your workload
	• Using write sharding to distribute workloads evenly
	  ○ Sharding using random suffixes
	  ○ Sharding using calculated suffixes
	• Distributing write activity efficiently during data upload
	• Best practices for using sort keys to organize data
	  ○ Using sort keys for version control
	• Best practices for using secondary indexes in DynamoDB
	  ○ General guidelines for secondary indexes in DynamoDB
	  ○ Take advantage of sparse indexes
	  ○ Using Global Secondary Indexes for materialized aggregation queries
	  ○ Overloading Global Secondary Indexes
		4 Hours (9th & 10th Day)

		○ Using Global Secondary Index write sharding for selective table queries ○ Using Global Secondary Indexes to create an eventually consistent replica • Best practices for storing large items and attributes ○ Compressing large attribute values ○ Storing large attribute values in Amazon S3 • Best practices for handling time series data in DynamoDB ○ Design pattern for time series data ○ Time series table examples • Best practices for managing many-to-many relationships ○ Adjacency list design pattern ○ Materialized graph pattern • Best practices for implementing a hybrid database system ○ If you don't want to migrate everything to DynamoDB ○ How a hybrid system can be implemented • Best practices for modeling relational data in DynamoDB ○ First steps for modeling relational data in DynamoDB ○ Example of modeling relational data in DynamoDB • Best practices for querying and scanning data ○ Performance considerations for scans ○ Avoiding sudden spikes in read activity ○ Taking advantage of parallel scans • Optimizing costs on DynamoDB tables ○ Evaluate your costs at the table level ○ Evaluate your table capacity mode	

	○ Evaluate your table's auto scaling settings ○ Evaluate your table class selection ○ Identifying your unused resources ○ Evaluate your Streams usage ○ Evaluate your provisioned capacity for right-sized provisioning	
AWS SDK	• Overview of AWS SDK support for DynamoDB • Working with DynamoDB using AWS SDK • Table-level Operations with AWS SDK • Write Operations - Item Level Operations with AWS SDK • Conditional Writes - Item Level Operations with AWS SDK • Atomic Counters - Item Level Operations with AWS SDK • Read Operations - Item Level Operations with AWS SDK • Paginated Reads - Item Level Operations with AWS SDK	**4 Hours (11th Day)**